SIVANA HAMER

Ph.D. Student in Computer Science | Researching Software Supply Chain Security

sahamer@ncsu.edu · sivanahamer.github.io · sivanahamer · sivanahamer · Sivana Hamer

Fourth-year Computer Science Ph.D. Candidate at North Carolina State University. I am currently researching the **state of software supply chain security** as a community **to help improve the security posture of industry and open-source projects**. I have published in top software engineering venues such as ICSE, TSE, FSE, and TOSEM. I am also a Google and Goodnight PhD fellow. I look forward to opportunities to conduct software supply chain security research.

RECENT EXPERIENCE

Graduate Research Assistant — Aug 2023–Present
North Carolina State University

Technical Intern — May 2026–Aug 2026
Applied Research Associates

Security Research Intern — May 2025–Aug 2025
Phoenix Security

FEATURED RESEARCH PROJECTS

Reduce Your Risk of Being Solarwinds, Log4j, or XZ Utils

- Analyzed the MITRE ATT&CK attack techniques in SolarWinds, Log4j, and XZ Utils to systematically synthesize software supply chain framework tasks to provide software organizations with a recommended starter kit of tasks. Collaboration with **Yahoo**.
- **Methods:** Qualitative Analysis, Incident Analysis, Meta Synthesis.
- **Results:** Frameworks are missing tasks; hence, even if all tasks were adopted, organizations would still be vulnerable to attacks.
- **Publication:** International Conference on Software Engineering.

Reputation Measures to Review Dependencies

- Investigated if network centrality measures, proxying contributor reputation, can be used as a signal to inform developers of dependency changes that require additional examination.
- **Methods:** Mixed-Methods, Statistical Models, Social Networks.
- **Results:** Network centrality measures are a significant factor in explaining how developers review dependencies in Rust.
- **Publication:** IEEE Transactions on Software Engineering.

Comparing Vulnerabilities ChatGPT and StackOverflow

- Compared the vulnerabilities of ChatGPT and StackOverflow to help raise software developers' awareness of the security implications when selecting code snippet platforms.
- **Method:** Quantitative Analysis, Statistical Methods.
- **Results:** ChatGPT generated less vulnerable code. Yet, insecure code propagation can happen in both platforms.
- **Tools:** Python, R, ChatGPT API, StackOverflow API, CodeQL.
- **Publication:** IEEE Security and Privacy Workshops.

RESEARCH INTERESTS

Software Measurement • Software Security • Empirical Software Engineering • System Security

EDUCATION

Ph.D. Computer Science
North Carolina State University

Aug 2023 – Present

Proposed Thesis: Automated Prevention of Software Supply Chain Compromises
Advisors: Dr. Laurie Williams & Dr. William Enck

M.Sc. Computer Science
Universidad de Costa Rica

2023

Advisor: Dr. Christian Quesada-López

B.Sc. Computer Science
Universidad de Costa Rica

2020

FEATURED AWARDS

- *Google PhD Fellowship (2025-2027).*
- *Goodnight Doctoral Fellowship (2023-2027).*
- *ICSE Distinguished Shadow Reviewer (2026).*
- *RSA Conference Security Scholar (2024).*
- *North Carolina State University Provost's Doctoral Fellowship (2023).*

SKILLS

- *Languages:* English, Spanish.
- *Programming languages:* Python, Java, R, C#, JavaScript, Bash, SQL.
- *Software tools:* Git, Jenkins, JIRA, Visual Studio Code, CodeQL, SonarQube, LLMs.
- *Frameworks and libraries:* ASP.NET, Flask, Bootstrap, jQuery, React, Unity, n8n.
- *Research methods:* Quantitative, Qualitative, Mining Software Repositories, Machine Learning, Statistical Models.

Last updated: November 2, 2026